

考える。

本量を

(2.1)

(2.2)

(2.3)

$$\begin{aligned} \delta &= E_{v^5} - \frac{G_{v^5}}{12} E_{v^4} + \left(\frac{7(G_{v^5})^2}{1200} - \frac{G_{v^6}}{30} \right) E_{v^3} \Big|_{v=0} \\ \hat{\delta} &= \hat{E}_{\eta^5} - \frac{\hat{G}_{\eta^5}}{12} \hat{E}_{\eta^4} + \left(\frac{7(\hat{G}_{\eta^5})^2}{1200} - \frac{\hat{G}_{\eta^6}}{30} \right) \hat{E}_{\eta^3} \Big|_{\eta=0} \\ &= \varepsilon E_{v^5} + 10 \varepsilon V_{\eta\eta} \textcircled{E_{v^4}} + (10 V_{\eta\eta\eta} + 15 \varepsilon \cancel{(V_{\eta\eta})^2}) \textcircled{E_{v^3}} \\ &\quad - \frac{1}{12} (\varepsilon G_{v^5} + 120 \varepsilon V_{\eta\eta}) (\textcircled{E_{v^4}} + 6 V_{\eta\eta} \textcircled{E_{v^3}}) \\ &\quad + \left\{ \frac{7}{1200} (G_{v^5} + 120 V_{\eta\eta})^2 - \frac{1}{30} (G_{v^6} + 27 V_{\eta\eta} G_{v^5} + 300 \varepsilon V_{\eta\eta\eta} + 1170 (V_{\eta\eta})^2) \right\} \varepsilon E_{v^3} \\ &= \varepsilon E_{v^5} + E_{v^4} \left(-\frac{\varepsilon}{12} G_{v^5} \right) + E_{v^3} \left[\frac{6 V_{\eta\eta} (-\frac{\varepsilon}{12} G_{v^5})}{+ \varepsilon \left\{ \frac{7}{1200} ((G_{v^5})^2 + 2 \cdot 120 \cdot V_{\eta\eta} G_{v^5}) - \frac{G_{v^6} + 27 V_{\eta\eta} G_{v^5}}{30} \right\}} \right] \\ &= \varepsilon \delta \end{aligned}$$

$$V_{\eta}(u,0) = \Sigma \in \{\pm 1\}$$

$$\frac{120 \cdot 120 \cdot 7}{1200}$$

$$\frac{15 - 10 \cdot 6}{+84 - 39} \varepsilon E_{v^3}$$

合計 n+1=37 の不変量	17	$K_S : E_{v^3}$	$\hat{K}_S = \varepsilon K_S$
	n=27	$\beta : E_{v^4}$	$\hat{\beta} = \beta$
	不変量	$\delta : E_{v^5}$	$\hat{\delta} = \varepsilon \delta$

・内的
・不変量!

$$\hat{G}_{v^5} : -\frac{\varepsilon}{2} V_{\eta\eta} + \frac{14\varepsilon}{10} V_{\eta\eta} - \frac{9}{10} V_{\eta\eta}$$

$$K: bdd \stackrel{?}{\iff} \beta = \delta = 0 \text{ on } U$$

$$K_S = E_V^3(u,0)$$

$$(K_\pi) \tilde{\pi}_0 \beta = E_V^4(u,0) - \frac{G_V^5(u,0)}{20} E_V^3(u,0)$$

$$\tilde{\pi}_0 \delta = E_V^5(u,0) - \frac{G_V^5}{12} E_V^4 + \left(\frac{7(G_V^5)^2}{1200} - \frac{G_V^6}{30} \right) E_V^3$$

$$\tilde{K} := v^2 K - \textcircled{*}$$

$$K = \frac{E(E_V G_V + G_U^2)}{4E^2 G^2} + \frac{G(E_U G_U + E_V^2)}{4E^2 G^2} - \frac{E_U + G_U}{2EG}$$

$$\begin{cases} E = 1 + \frac{1}{3!} E_V^3(u,0) v^3 + \frac{1}{4!} E_V^4(u,0) v^4 + \frac{1}{5!} E_V^5(u,0) v^5 + e_6(u,v) v^6 \\ G = \frac{1}{4} v^4 + \frac{1}{5!} G_V^5(u,0) v^5 + \frac{1}{6!} G_V^6(u,0) v^6 + g_7(u,v) v^7 \end{cases}$$

これを $\textcircled{*}$ に代入

$$\tilde{K} = \underbrace{u \text{ の関数}}_{\beta(u)} + \underbrace{u \text{ の関数}}_{\delta(u), \beta(u)} v + \exists k(u,v) v^2$$

$\therefore K : \text{bdd}$

$$\Leftrightarrow \square = \bigcirc = 0$$

$\pi_{3,4} > 0$



$n=1$ (ただし $\textcircled{0}$)

K_S

K_U

K_π

$\left(\begin{matrix} \omega_{2,3} \\ \pi_{2,3} \end{matrix} \right) \leftarrow \text{MSSTZ}$
 $(= K_V \omega_{2,3})$
 $K_\pi^{2,3} ? \pi_{2,3}$
 $\uparrow$
 $\downarrow$ δ

to do $\textcircled{HW}$

• $\beta(u)$ と $\pi_{3,4}(u)$ の関係

$$(\pi_{3,4}(u) = -\frac{1}{3} \beta(u))$$

• $\delta(u)$ と $\pi_{3,5}(u)$ の関係

($\beta(u) \equiv 0$ のとき
 δ と $\pi_{3,5}$ は定数倍一致?)

• $\pi_{3,5}(u)$ は内在的か?

• $\beta \equiv 0$ のときは内在的か?

• $\beta \neq 0$ のときは外在的?
 もしそうなら HMM を
 使ってみよう?

$$\delta = E_V^5 - \frac{G_V^5}{12} E_V^4 +$$

$$\hat{\delta} = \hat{E}_V^5 - \frac{\hat{G}_V^5}{12} \hat{E}_V^4 +$$

$$= \varepsilon E_V^5 + 10 \varepsilon V_{\eta\eta}$$

$$- \frac{1}{12} (\varepsilon G_V^5 + 12$$

$$+ \left\{ \frac{7}{1200} \right.$$

$$= \varepsilon E_V^5 + E_V^4$$

$$= \varepsilon \delta$$

合計 $\left[\begin{matrix} n+1=37 \\ \text{の不变量} \end{matrix} \right] \rightarrow K_S$
 $\left[\begin{matrix} n=27 \\ \text{の不变量} \end{matrix} \right]$

$$\pi_{3,4} > 0.$$

$n=1$ (1220 @)

$$\begin{aligned} K_S &= W_{2,3} \leftarrow \text{MSSTE} \\ K_L &= \pi_{2,3} (= K_V W_{2,3}) \end{aligned}$$

to do

HW

- $\beta(u)$ と $\pi_{3,4}(u)$ の関係
(? $\pi_{3,4}(u) = -\frac{1}{3}\beta(u)$)
- $\delta(u)$ と $\pi_{3,5}(u)$ の関係
(? $\beta(u) \equiv 0$ のとき
 δ と $\pi_{3,5}$ は定数倍一致?)
- $\pi_{3,5}(u)$ は内在的か?
• $\beta \equiv 0$ のときは内在的左3
• $\beta \neq 0$ のときは外在的?
もし左3なら HMM を
使って示せる?

$$\delta = E_{V^5} - \frac{G_{V^5}}{12} E_{\beta^4} + \left(\frac{7(G_{V^5})^2}{1200} - \frac{G_{V^6}}{30} \right) E_{V^3} \Big|_{V=0}$$

$$\hat{\delta} = \hat{E}_{\eta^5} - \frac{\hat{G}_{\eta^5}}{12} \hat{E}_{\eta^4} + \left(\frac{7(\hat{G}_{\eta^5})^2}{1200} - \frac{\hat{G}_{\eta^6}}{30} \right) \hat{E}_{\eta^3} \Big|_{\eta=0}$$

$$= \varepsilon E_{V^5} + 10 \varepsilon V_{\eta\eta} E_{V^4} + (10 V_{\eta\eta\eta} + 15 \varepsilon (V_{\eta\eta})^2) E_{V^3}$$

$$- \frac{1}{12} (\varepsilon G_{V^5} + 120 \varepsilon V_{\eta\eta}) (E_{V^4} + 6 V_{\eta\eta} E_{V^3})$$

$$+ \left\{ \frac{7}{1200} (G_{V^5} + 120 V_{\eta\eta})^2 - \frac{1}{30} (G_{V^6} + 27 V_{\eta\eta} G_{V^5} + 300 \varepsilon V_{\eta\eta\eta} + 1170 (V_{\eta\eta})^2) \right\} E_{V^3}$$

$$= \varepsilon E_{V^5} + E_{V^4} \left(-\frac{\varepsilon}{12} G_{V^5} \right) + E_{V^3} \left[6 V_{\eta\eta} \left(-\frac{\varepsilon}{12} G_{V^5} \right) + \varepsilon \left\{ \frac{7}{1200} ((G_{V^5})^2 + 2 \cdot 120 \cdot V_{\eta\eta} G_{V^5}) - \frac{G_{V^6} + 27 V_{\eta\eta} G_{V^5}}{30} \right\} \right]$$

$$= \varepsilon \delta$$

$$V_{\eta}(u,0) = \varepsilon \in \{\pm 1\}$$

$$\frac{120 \cdot 120 \cdot 7}{1200}$$

$$\frac{15 - 10 \cdot 6 + 84 - 39}{\varepsilon E_{V^3}}$$

合計 $n+1=37$ の不変量	$n=1 \rightarrow K_S : E_{V^3}$ $n=27$ 不変量 $\beta : E_{V^4}$ $\delta : E_{V^5}$	$\hat{K}_S = \varepsilon K_S$ $\hat{\beta} = \beta$ $\hat{\delta} = \varepsilon \delta$
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内
不
変
量!

$$G_{V^5} : -\frac{\varepsilon}{2} V_{\eta\eta} + \frac{14\varepsilon}{10} V_{\eta\eta} - \frac{9}{10} V_{\eta\eta}$$

$$K: \text{bdd} \Rightarrow \beta = \delta = 0 \text{ on } U\text{-軸}$$

$$K_S = E_V(u,0)$$

$$(K_S)_{\tilde{H}_0}[\beta] = E_V(u,0) - \frac{G_V(u,0)}{20} E_V(u,0)$$

$$\tilde{\delta} = E_V(u,0) - \frac{G_V}{12} E_V + \left(\frac{7(G_V)^2}{1200} - \frac{G_V}{30} \right) E_V$$

$$\tilde{K} := v^2 K - \otimes$$

$$K = \frac{E(E_V G_V + G_V^2)}{4E^2 G^2} + \frac{G(E_V G_V + E_V^2)}{4E^2 G^2} - \frac{E_V + G_V}{2EG}$$

$$\begin{cases} E = 1 + \frac{1}{3!} E_V(u,0) v^3 + \frac{1}{4!} E_V(u,0) v^4 + \frac{1}{5!} E_V(u,0) v^5 + e_6(u,v) v^6 \\ G = \frac{1}{4} v^4 + \frac{1}{5!} G_V(u,0) v^5 + \frac{1}{6!} G_V(u,0) v^6 + g_7(u,v) v^7 \end{cases}$$

これを(*)に代入

$$\tilde{K} = \underbrace{u \text{ の関数}}_{\beta(u)} + \underbrace{u \text{ の関数}}_{\delta(u), \beta(u)} v + k(u,v) v^2$$

(HW) (3.4)-カスゾロ
(3.5)-カスゾロ
(Hartman, §9.8)
の判定条件
を証明



n=1 (カスゾロ)
 $K_S = \omega_{2,3} \leftarrow \text{MSSTE}$
 $K_C = \omega_{2,3}$
 $K_\pi = \pi_{2,3} (= K_V \omega_{2,3})$
 $K_\pi^{2,3} ? \pi_{2,3}$
 $\uparrow$
 1.2.3

to do (HW)

- $\beta(u)$ と $\pi_{3,4}(u)$ の関係
(? $\pi_{3,4}(u) = -\frac{1}{3} \beta(u)$)
- $\delta(u)$ と $\pi_{1,5}(u)$ の関係
(? $\beta(u) \equiv 0$ のとき
 δ と $\pi_{1,5}$ は定数倍一致?)
- $\pi_{3,5}(u)$ は内在的か?
 • $\beta \equiv 0$ のときは内在的
 • $\beta \neq 0$ のときは外在的?
 もしそうなら HMM を
 使ってみよう?

$$\begin{aligned} \delta &= E_V - \frac{G_V}{12} E_V + \left(\frac{7(G_V)^2}{1200} - \frac{G_V}{30} \right) E_V \\ \hat{\delta} &= \hat{E}_V - \frac{\hat{G}_V}{12} \hat{E}_V + \left(\frac{7(\hat{G}_V)^2}{1200} - \frac{\hat{G}_V}{30} \right) \hat{E}_V \\ &= \varepsilon E_V + 10 \varepsilon V_{\eta\eta} E_V + \left(\frac{7}{12} (\varepsilon G_V + 120 \varepsilon V_{\eta\eta}) \right. \\ &\quad \left. - \frac{1}{12} (\varepsilon G_V + 120 \varepsilon V_{\eta\eta}) \right) E_V \\ &= \varepsilon E_V + E_V \left(-\frac{\varepsilon}{12} G_V \right) \\ &= \varepsilon \delta \end{aligned}$$

合計
 $n+1=37$
 の不変量

$17 \rightarrow K_S : E_V$	$\left(\begin{array}{l} \beta : E_V^4 \\ \delta : E_V^5 \end{array} \right)$
$n=27$	

$$K_S = E_V(u,0)$$

$$(K_{\pi})_{\pi\pi} \beta = E_V(u,0) - \frac{G_V^5(u,0)}{20} E_V(u,0)$$

$$\delta = E_V(u,0) - \frac{G_V^5}{12} E_V + \left(\frac{7(G_V^5)^2}{1200} - \frac{G_V^6}{30} \right) E_V$$

$$\tilde{K} := v^2 K \quad \text{---} \textcircled{*}$$

$$K = \frac{E(E_V G_V + G_V^5)}{4E^2 G^2} + \frac{G(E_V G_V + E_V^2)}{4E^2 G^2} - \frac{E_V + G_V}{2EG}$$

$$\begin{cases} E = 1 + \frac{1}{3!} E_V^3(u,0) v^3 + \frac{1}{4!} E_V^4(u,0) v^4 + \frac{1}{5!} E_V^5(u,0) v^5 + e_6(u,v) v^6 \\ G = \frac{1}{4} v^4 + \frac{1}{5!} G_V^5(u,0) v^5 + \frac{1}{6!} G_V^6(u,0) v^6 + g_7(u,v) v^7 \end{cases}$$

これを $\textcircled{*}$ に代入

$$\textcircled{\text{HW}} \quad \tilde{K} = \boxed{\text{uの関数}} + \textcircled{\text{uの関数}} v + \exists k(u,v) v^2$$

$\uparrow$ $\beta(u)$ $\uparrow$ $\delta(u), \beta(u)$

$\textcircled{\text{HW}}$ (3.4)-カスワ①
(3.5)-カスワ②) の判定条件
(Hattori, §9.8)
を参照



$n=1$ (カスワ①)
 $K_S = \textcircled{W_{2,3}} \leftarrow \text{MSSTE}$
 $K_C = \textcircled{W_{2,3}}$
 $K_{\pi} = \textcircled{\Pi_{2,3}} (= K_V W_{2,3})$
 $\uparrow$ $K_{\pi}^{2,3} ? \Pi_{2,3}$
 $\uparrow$ $\textcircled{\text{1.2.3}}$

to do $\textcircled{\text{HW}}$

- $\beta(u)$ と $\Pi_{3,4}(u)$ の関係
 $(? \Pi_{3,4}(u) = -\frac{1}{3} \beta(u))$
- $\delta(u)$ と $\Pi_{3,5}(u)$ の関係
 $(? \beta(u) \equiv 0 \text{ とき } \delta \text{ と } \Pi_{3,5} \text{ は定数倍一致?})$
- $\Pi_{3,5}(u)$ は内在的か?
 $\bullet \beta \equiv 0$ のときは内在的
 $\bullet \beta \neq 0$ のときは外在的?
 もし外在的なら HMM を使って示せよう?

$$\begin{aligned} \delta &= E_V^5 - \frac{G_V^5}{12} E_V + \left(\frac{7(G_V^5)^2}{1200} - \frac{G_V^6}{30} \right) E_V \\ \hat{\delta} &= \hat{E}_V^5 - \frac{\hat{G}_V^5}{12} \hat{E}_V + \left(\frac{7(\hat{G}_V^5)^2}{1200} - \frac{\hat{G}_V^6}{30} \right) \hat{E}_V \\ &= \varepsilon E_V^5 + 10 \varepsilon V_{\eta\eta} \textcircled{E_V^5} + (10 \varepsilon G_V^5 - \frac{1}{12} (\varepsilon G_V^5 + 120 \varepsilon V_{\eta\eta})) E_V \\ &\quad + \left\{ \frac{7}{1200} (G_V^5 + 120 V_{\eta\eta})^2 - \frac{G_V^6}{30} \right\} E_V \\ &= \varepsilon E_V^5 + E_V^5 \left(-\frac{\varepsilon}{12} G_V^5 \right) \\ &= \varepsilon \delta \end{aligned}$$

合計 $n+1=37$ の不変量

$17 \rightarrow K_S : E_V^3$	$\hat{K}_S$ $\hat{\beta}$ $\hat{\delta}$
$n=27$ 不変量 $\left(\begin{matrix} \beta : E_V^4 \\ \delta : E_V^5 \end{matrix} \right)$	